Calculator Review and Revision Log

Revisit 1 Activity Data, Emission Factors and Calculation check. Prepare an evidence-based revision of Prototype 0.

Group: 9	Date: 25 October 2026	Prototype 0 filename: Group9_Prototype0.html

1 Map the activity data

List every activity input required by Prototype 0. Match it to the Shared Student Profile. Identifying incomplete or missing information.

Profile status: **Provided / Partial / Not provided**. For a gap, choose a reasonable and transparent approach to handle it. A proxy or default should be documented and editable; unknown information should not be presented as fact. Add rows if needed.

Category and input in P0	Required unit and time period	Shared Profile status and available information	Proposed treatment of the data gap	Calculator support or revision needed
Transport: walking or cycling on campus	km/week	Provided. About 20 km/week.	Use the shared value as the default.	None. Prototype 0 already does this.
Transport: metro trips and one-way distance	trips/week and km/trip	Provided. 2 one-way trips per week, 15 km each.	Use both shared values. Weekly distance is converted to the teaching semester.	None.
Transport: ride-hailing trips and one-way distance	trips/month and km/trip	Provided. 2 one-way trips per month, 10 km each, usually alone.	Count each trip as one passenger. Do not divide by an assumed group.	None. No occupancy input was added.
Transport: high-speed rail round trips and one-way distance	round trips/teaching semester and km one way	Provided. 2 round trips per teaching semester, about 220 km one way.	Keep the shared values. One round trip is two one-way distances.	None for the activity data. The emission factor is revised in section 4.
Energy: room electricity	kWh/last 30 days, whole room	Partial. 120 kWh is the whole room. Personal use is not given, and 30 days is not a semester.	Keep 120 kWh as the editable room total. State two assumptions: split the room equally among 4 people, and scale 30 days to 126 days because one semester is assumed to be 18 weeks.	None. Both assumptions are already shown. Eighteen weeks is not a shared fact.
Energy: hot-shower count and duration	times/week and minutes/shower	Partial. 6 times per week, about 8 minutes each. Energy source and amount are not provided.	Record the count and duration. Leave them out of the total and state why. Do not invent heater power or water flow.	None. Prototype 0 already records and excludes them.
Food: four main-meal counts	meals/week	Provided. Beef or lamb 3, other meat 8, fish/eggs/dairy/soy 5, mostly plant-based 5. About 2 takeaway meals per week are already inside these 21 meals.	Use the four counts only. Do not add a separate takeaway-meal input.	None.
Shopping: online parcels and clothing	parcels/month and items/teaching semester	Partial. 3 parcels per month and 2 clothing items per semester. Material, price, packaging weight, and lifetime are not provided.	Record the counts. Leave them out of the total and state why.	None.
Waste: general waste	small bags/week	Partial. About 3 small bags per week. Mass, treatment pathway, and whether	Record the bag count. Do not divide by 4. Leave it out of the total and state why.	None.

Category and input in P0	Required unit and time period	Shared Profile status and available information	Proposed treatment of the data gap	Calculator support or revision needed
		the bags are personal or for the whole room are not provided.		
Waste: recyclable bottles or cans	items/week	Partial. About 3 per week. Mass and treatment pathway are not provided.	Record the count. Leave it out of the total and state why.	None.

For partial or missing data, decide what support the calculator should provide. It may guide the user to obtain the information, offer a transparent and editable estimation method, allow N/A or skip when a reasonable estimate is not possible, and explain how the estimate or omission affects the result. Do not silently treat missing data as zero or use an undocumented default.

Example: The Shared Profile provides Li Ming's shower frequency and duration, but not the hot-water energy source or energy consumption. The calculator could ask the user to identify the energy source, offer an "I don't know" option, and provide a transparent estimate only when its assumptions are clearly stated and editable.

2 Review emission factors

Review at least three emission factors from at least two categories. For each factor, identify the source, locate the original document or webpage, and confirm that the value and unit used in the calculator match the source. For this submission, record three factors; before the final calculator is submitted, every factor used should have an identifiable source.

Activity and category	Factor currently used	Source check	Fit with this calculation	Decision
EF 1 Corresponding activity: dorm electricity used by Li Ming Category: Energy	Value and unit: 0.57 kg CO2e/kWh	Source shown in P0: No Original source can be located: Yes The number 0.57 itself is not in a cited document. The located factor is the 2023 Shanghai provincial average.	Compatible with the activity-data unit: Yes Activity data are in kWh. Relevant geography and year: Shanghai, 2023. The Prototype 0 value has no stated geography or year.	Replace Replacement: 0.5737 kg CO2/kWh. Ministry of Ecology and Environment and National Bureau of Statistics, Announcement on the 2023 Electricity Carbon Dioxide Emission Factors (2025, No. 47), Shanghai provincial average, for location-based electricity consumption. The source reports CO2, not a full CO2e factor. https://www.mee.gov.cn/xxgk2018/xxgk/xxgk01/202512/W020251231726284332528.pdf
EF 2 Corresponding activity: high-speed rail home Category: Transport	Value and unit: 0.03 kg CO2e/passenger-km	Source shown in P0: No Original source can be located: Yes, for the replacement below. The number 0.03 itself was not found in a source.	Compatible with the activity-data unit: Yes Activity data are passenger-kilometres. Relevant geography and year: China, 2019, operation phase.	Replace Replacement: 0.02709 kg CO2/passenger-km (27.09 g CO2/passenger-km). Tong Ruiyong, Wei Runbin, Wu Jinyan, Mao Baohua, and Tian Peining, 2025. Influence of power generation structure on carbon emission factor of high-speed railway in operation period. Climate Change Research, 21(1): 116-124. Includes upstream electricity generation. The source reports CO2. https://www.climatechange.cn/EN/Y2025/V21/I1/116

Activity and category	Factor currently used	Source check	Fit with this calculation	Decision
EF 3 Corresponding activity: beef or lamb main meals Category: Food	Value and unit: 6 kg CO2e/meal	Source shown in P0: No Original source can be located: No	Compatible with the activity-data unit: Unclear Published food factors are usually per kilogram of food. The shared profile gives meals, not meal weight. Relevant geography and year: not identified.	Cannot verify yet No replacement in this revision. The per-meal factor stays marked as not yet verified.

3 Check the calculator

Use the interface to look for calculation or category errors. You do not need to inspect the code or repeat every multiplication by hand.

Check	Input setting	What should happen	What actually happened	Issue and revision needed
Single activity	Set every editable activity input to zero. Enter only room electricity: 120 kWh over the last 30 days.	Only the relevant category and total should show emissions attributable to that input.	Energy and the total both showed 71.8 kg CO2e per teaching semester. Transport and food showed 0.0. A second run with only metro (2 trips/week, 15 km/trip) showed 21.6 kg in transport and in the total, and 0 elsewhere.	No category error. Walking stays at 0 because its use-phase factor is 0. Parcels, clothing, showers, and waste do not move the total because they are recorded only. That matches the stated rules. The unsourced factor values are handled in section 2, not by this arithmetic check.
Change one input	Start from Li Ming's defaults. Change beef or lamb from 3 to 4 meals/week. Leave every other input unchanged.	The relevant category and total should change in the expected direction; unrelated categories should not change.	Food rose by 108 kg, from 648.0 to 756.0. The total rose from 785.8 to 893.8 kg CO2e per teaching semester. Transport stayed 66.0 and energy stayed 71.8.	No arithmetic revision. Unrelated categories did not change. Electricity and high-speed rail are revised in section 4 because their factors had no source.

4 Prepare the revision brief

Include only decisions that require a calculator to be revised. Entering Li Ming's data is user work unless the calculator needs conversion, missing-data treatment or another design change.

Finding or issue	Evidence or approved rule	Specific calculator revision	Priority
The electricity factor in Prototype 0 has no source.	Ministry of Ecology and Environment and National Bureau of Statistics, 2025 Announcement No. 47. The 2023 Shanghai provincial average is 0.5737 kg CO2/kWh and matches kWh of dorm electricity.	Replace 0.57 kg CO2e/kWh with 0.5737 kg CO2/kWh and cite the announcement in the factor table.	High
The high-speed rail factor in Prototype 0 has no source.	Tong et al., Climate Change Research, 2025, 21(1): 116-124. China high-speed rail, 2019	Replace 0.03 kg CO2e/passenger-km with 0.02709 kg CO2/passenger-km and cite the paper in the factor table.	High

Finding or issue	Evidence or approved rule	Specific calculator revision	Priority
	operation phase: 27.09 g CO2/passenger-km.		
Prototype 0 labels every factor as a placeholder.	After two factors have sources, a blanket placeholder label would misstate them. Factors that are still unsourced must stay visibly separate.	Add status and source columns. Mark electricity and high-speed rail as sourced. Mark the others as not yet verified. State beside the total that the two sourced factors are CO2, so the semester total is not a complete CO2e figure.	High

Unresolved information or limitations to retain in Prototype 1.0: Metro (0.04), ride-hailing (0.20), walking (0), and all four meal factors remain unverified. A 2012 Shanghai metro figure was not adopted because the electricity factor embedded in it does not match the 2023 Shanghai grid factor. Showers, parcels, clothing, waste, and takeaway packaging stay outside the total. The 18-week semester, the 4-week month, and the equal 4-person electricity split remain labeled assumptions. Shared-profile defaults are unchanged.